

Relations among Perceived Environmental Uncertainty, Organization Structure, and Boundary-Spanning Behavior

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Boundary-spanning activity is conceived of as an intervening variable between organization structure and perceived environmental uncertainty (PEU). This is due primarily because, since boundary spanners operate at the skin of the organization, their functions are to interpret environmental conditions and relay that information to organization decision makers. Three hypotheses examining relations between PEU, organization structure, and boundary-spanning activity were investigated. In a field study comprising 12 work units (182 people) working in a health and welfare organization, positive relations were found between all three variables. Partial relations indicated that PEU does not influence the relationship of boundary spanning to structure but structure reduces the relationship of PEU to boundary spanning to zero, and boundary spanning reduces the relationship between structure and PEU to zero. Thus, not only is the notion of boundary spanning as an intervening variable supported, but results suggest that causality in the relationship between structure and PEU may be from structure to PEU rather than current contingency notions of PEU affecting structure. •

This paper reports the results of a field study aimed at delineating the relationships among perceived environmental uncertainty (PEU), organization structure, and boundary-spanning behavior. Although a number of scholars have studied the relationship between an organization's environment and its structure (see Downey and Slocum, 1975; Darran, Miles, and Snow, 1975; for reviews of this literature), there have apparently been few if any studies of the possibility that these variables or this relationship might be related to the boundary-spanning behavior of the organization's members.

Organizational boundaries are important both because they function to circumscribe or limit organizations (see Child, 1972; Evan, 1966; Levine and White, 1961; Utterback, 1971), and because they function as regulators of the flow of information between the organization and its environment (see Miller, 1972; Miller and Rice, 1967; Organ, 1971; Thompson, 1967). Recent work by Adams (1976), Aldrich and Herker (1977), Miles (1976), and Keller, Szilagyi, and Holland (1976) all point to the importance of the activities of those organizational members who operate at the organizational boundaries, boundary spanners, for organizational functioning. In view of the potential importance of this boundary-crossing information flow to a more complete theory of organizations and their environment, this study seeks to determine whether the frequency of boundary-spanning activity is related to either PEU or organization structure.

This study is concerned with the uncertainty in the environment that is perceived by the organization member. Miles, Snow, and Pfeffer (1974) in particular suggested the appropriateness of using this variable in empirical studies of open systems or contingency theory, that is, the investigation of differential influences or contingencies in the

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environment on organizational structures. These researchers found that, within the same objective environment, there were organizations (college textbook publishers, in this case) whose top managers perceived little or no change and uncertainty in the environment as well as organizations whose top managers perceived continuous change and uncertainty in the environment. They then noted (1974: 262) that these two types of perceptions, and hence the organizational structures that were associated with them, can survive together and even flourish, at least in the short run. This reasoning, the review and conceptual work of Downey and Slocum (1975) and Galbraith (1973), and also Weick's notion of an "enacted environment" (1969) have all suggested that the organizational design actions an organization takes in responding to its environment may well be more consistent with perceptions of the environment than with more objective indicators of environmental conditions. Hence, in this study, the concept of organizational environment was operationalized as perceived environmental uncertainty or PEU.

In particular, the study addresses two related issues. The first is whether or not "contingency theory," as the term is currently used in discussions of the organization-environment interface (see Kast and Rosenzweig, 1970; Utterback, 1971), is deficient because it does not include as a variable the boundary-spanning activity that characterizes this interface. Since the theory focuses so strongly on the concept of information passing from the environment to the organizational designers, it may be important to consider including in the theory a variable dealing with the intensity of information-gathering activity, a variable that may mediate the relationship between the organization and its environment. The second issue addressed in the study concerns the linkages that might exist among organizational structure, boundary-spanning behavior, and PEU. Although significant correlations may be observed between any two of these variables, they may be simply consequences of correlations with the third variable and not indicative of any underlying causal relation. Thus there are two issues—one concerning the need to add a variable to an existing theory, and the other concerning the first- and second-order relations among the three variables of this study.

Hierarchical Level and Information Processing

The data for this study were obtained in the lowest two echelons of a rather tall, state government agency. Information gathered by such echelons has been recognized as an integral part of organizational decision making. For example, Brown (1966) elaborated on Parson's (1960) taxonomy of decision-making levels (that is, the institutional level, the managerial level, and the technical level), by describing the kinds of environmental information necessary at each level for appropriate decision making. In particular, Brown suggested that the technical level (the lowest echelon) intelligence gathering mechanism provides operating information by interacting with the environment at the level of suppliers, customers, and competitors.

Because such information has an effect on the decision makers' PEU, and consequently on his or her decisions, an analysis of boundary spanning at the lowest organizational levels is necessary to understand organizational information processing.

HYPOTHESES

Three hypotheses are tested in this study, each one describing the relation between two of the three variables studied. Each hypothesis is tested using both first- and second-order analyses.

The first hypothesis concerns the relationship between perceived environmental uncertainty and the nature of the organizational structure. Contingency theory suggests that PEU affects organizational structure, that administrators design organizational structures such that the organization will be able to more effectively respond to environmental demands, and that they do this in accord with their perceptions of the environment. In general, it is argued that the more uncertainty that is perceived, the more will be the looseness or flexibility or "organicness" of the organizational structure, that is, that a positive relationship exists between these variables. For example, on the basis of a number of field studies Osborn and Hunt (1974) concluded that either perceived or objective environmental uncertainty is a determinant of organizational structure. In particular they stated (1974: 232):

The work of Burns and Stalker (1961), Chandler (1962), Emery and Trist (1965), Lawrence and Lorsch (1967), and Neghandi and Reiman (1973), among others, has indicated that as [the task environment] becomes more dynamic, the organization must become not only more receptive to change, but alter its internal structure and operations to maintain and/or establish a high survival potential.

If various structures are differentially effective in processing information to reduce decision-making uncertainty, then the information processing capabilities of structure would need to be consistent with the uncertainty perceived in the environment. The field studies of Duncan (1971), using data from 22 decision groups, and Khandwalla (1972), using data from 29 manufacturing firms, both found that increased perceived uncertainty was correlated with less mechanistic structures for effective organizations, thus apparently supporting the PEU influences structure argument. McWhinney (1968) used the same argument in proposing various decision-making modalities appropriate to the types of environments described by Emery and Trist (1965).

In this research, structure is conceptualized on a mechanistic-organic continuum, and the adjective "organicness" indicates the extent to which an organization possesses what Burns and Stalker (1966) would call an organic structure, that is, a structure not characterized by highly specified and delimited job descriptions or highly formalized procedures.

On the basis of the above arguments the following is hypothesized:

Hypothesis 1. Organicness of structure will be positively associated with PEU.

While the literature, in general, suggests an uncertainty-organicness fit for total institutions, a number of writers, particularly Thompson (1967) view organizations as composed of both open and buffered components.

The effect of this buffering on the relation between structure and PEU may be considerable, as suggested in the experimental study by Huber, O'Connell, and Cummings (1975). In that study, holding objective environmental uncertainty constant, they found that organicness was negatively associated with PEU. One explanation for this is that, in the highly structured inorganic groups of their laboratory, the buffering was a result of rather extreme job specialization and the increase in uncertainty created by communication restriction.

Since this laboratory setting was not characteristic of many field settings and certainly not of the one to be described in the methodology section of this research, the Huber, O'Connell, and Cummings (1975) finding was not used in the development of hypothesis 1.

The second hypothesis concerns the relationship between the frequency of the boundary-spanning activity and the nature of the organizational structure. Except for the work of Aiken and Hage (1972), this relationship is empirically unstudied. They found (1972: 5):

... the greater the degree of organizational permeability, i.e., the greater the extensity of boundary-spanning roles and the greater the number of interorganizational linkages, the greater the internal structure of the focal organization will tend toward the organic (Burns and Stalker, 1961) or dynamic (Hage and Aiken, 1970) organizational style.

Arguments from the systems theory literature (Miller, 1972) and cybernetics (Ashby, 1968) suggested that openness of a system's boundaries increased the amount of boundary activity. Openness is associated with richer connections between systems components and nonsystems components. This increased connectedness implies less control by intrasystem elements (that is, less structure). Thus, Duncan (1971) predicted that, because in more certain environments information processing requirements will be minimal, the organization will rely more on established rules and procedures, and would require less boundary spanning. Hypothesis 2 is developed from an elaboration on their statement. In uncertain environments, organizations will have a high need (1) to obtain information, for example, to have a high frequency of boundary-spanning activity, and (2) to process it in a way that will maximize the number of points of view to which it is exposed, for example, to create an open system or organic structure, and that in more certain environments, organizations will have a lower need to do either. If these arguments hold, then it is clear that boundary-spanning activity and organicness would be associated with one another. As a consequence then, hypothesis 2 can be stated.

Hypothesis 2. Organicness will be positively associated with frequency of boundary-spanning activity.

The third hypothesis concerns the relationship between PEU and boundary spanning. That is, if organizational members attempt to reduce perceived environmental un-

certainty by obtaining more information, it follows that those having high PEU would engage in more boundary-spanning activity in order to bring that uncertainty down to some manageable level. Similarly, in environments that are not perceived as uncertain, one would expect to find low perceived uncertainty and also less boundary-spanning activity. Thus, the third hypothesis is as follows.

Hypothesis 3. Frequency of boundary-spanning activity will be positively associated with PEU.

METHODOLOGY

This cross-sectional field study was conducted in a health and welfare organization, concerned with family problems, adoption, social work, and so forth, in a midwestern state government. Data were obtained by administering a questionnaire to all 220 employees in 12 work groups. Each work group was semi-autonomous and included clerical, managerial, and caseworker personnel. Each questionnaire was coded to provide anonymity of respondents. A member of the organization assigned a number to each name. Completed questionnaires were given directly to the research team. Thus the agency did not see individual results and the researcher did not see the names. The questionnaire was administered to groups of no more than 15 persons. The organization assigned personnel to keep track of absences. Follow-up testing of absentees consisted of giving each one a questionnaire to be returned by mail. One hundred eighty-two questionnaires were obtained for a response rate of 80 percent.

Measures of the Variables

Three composite variables are used in the field study: organicness, PEU, and frequency of boundary-spanning activity.

Six measures were included in the organicness variable.¹ Five of these, (1) participation in work decisions, (2) formalization (reverse scored), (3) hierarchy of authority (reverse scored), (4) impersonality (reverse scored), and (5) division of labor measures are taken from Duncan (1971). The sixth measure is a questionnaire item directed at ascertaining the extent of the subject's participation in strategic decisions. Spearman-Brown reliabilities are: hierarchy of authority, .91; impersonality in decision making, .72; rules and procedures, .74; division of labor, .65; participation in strategic decisions, .81; participation in work decision making, .78. Table 1 compares the measures of structure used here with those attributes of an organic system as defined by Burns and Stalker (1966).

The proposed measures were factor analyzed to determine if all of them merited inclusion in the variable. The criterion for excluding a measure was that it would have little association with (loading on) the first factor. This type of assessment was made for the PEU and boundary-spanning variables as well as for the organicness variable.

With respect to the organicness variable, the mean score across the relevant questions of the structure instrument was used as the score for each measure. The results of

1
The questionnaire and its development are contained in Leifer (1975). Copies of the questionnaire are available upon request.

Table 1

Comparison of Structural Variables Used in This Research with Those Used to Describe Organic Structures

Present Research Developed by Duncan (1971)		Organic Management System Burns and Stalker (1966)
Division of labor (Specialization)	Low	1. Specialized differentiation of each individual task (mechanistic)
	High	2. The contributive nature to the common task, realistic nature of the individual task (organic)
Impersonality	High	1. Communication governed by instructions and decisions issued by superiors (mechanistic)
	Low	2. Content of communication information and advice rather than instructions, resembling consultation (organic)
Participation in decision making	Low	1. Hierarchic structure by location of knowledge at top of hierarchy where final reconciliation and assessment is made (mechanistic)
Centralization	High	2. Ad hoc center of control authority and communications located anywhere in the network based on knowledge about the task (organic)
Hierarchy of authority	High	1. Hierarchical structure of control authority and communication (mechanistic)
	Low	2. Control, authority, and communication derive from community of interest with the rest of the organization (organic)
Formalization	High	1. Precise definition of . . . each role (mechanistic)
	Low	2. The adjustment and continual redefinition of individual tasks through interaction with others (organic)

factor analyzing these six scores are shown in Table 2. Because the factor loadings were all high, all six measures were included in the variable. The unweighted mean scores on each measure were then averaged to obtain an organicness score for each individual's work unit. The average of these organicness scores for all individuals in a unit was used as the organicness score for that unit. These unit organicness scores ranged from 38 to 72 on a 0 to 100 scale, and had a mean of 58 and a median of 57.

PEU was measured by seven questionnaire items of the form, "How often do you know what to expect in your dealings with other people or organizations?"² The responses were collected on a 0 to 100 scale anchored at five points; never, seldom, sometimes, often, and always. A factor analysis indicated that each question loaded on the first factor with a loading of from .37 to .75 (Table 3 contains the factor matrix and eigenvalues). The factor loadings were sufficiently positive so that all seven questions were used as measures of the composite variable. These seven unweighted scores were then added together and their mean value used as the PEU score for each individual. These PEU scores ranged from 34 to 93, with a mean of 42 and a median of 41 on a 0 to 100 scale.

² The questionnaire and its development are contained in Leifer (1975). Copies of the questionnaire are available upon request.

Table 2

Factor Matrix for Structure Measures			
Item	Factor Matrix		<i>h</i> ²
	1	2	
Participation in work decisions	.60	−.31	.45
Participation in strategic decisions	.86	−.22	.79
Hierarchy of authority (reverse scored)	.82	−.26	.74
Impersonality (reverse scored)	.80	−.15	.66
Formalization (reverse scored)	.54	.63	.69
Division of labor	.59	.62	.73
Eigenvalues	3.04	1.02	

In this study, boundary-spanning activity is defined by the total verbal and written interactions with extra work units. The questions shown in Table 4 were used to create a composite boundary-spanning variable. A factor analysis indicated that each of the eight questions loaded on the first factor with a loading of from .44 to .83 (Table 5 contains the factor matrix and eigenvalues). These eight unweighted scores were added together and their sum used as the boundary-spanning activity score for each individual. These values ranged from 0 to 242 with a median score of 26 and an average score of 46.

It should be noted that this composite measure of boundary-spanning activity, based as it is on behavior, is conceptually quite different from the composite measure of organizational structure described earlier. Close inspection of the eight boundary-spanning activity measures and the six organicness measures suggests that the frequencies of the eight boundary-spanning behaviors are determined by the motivations and, to a lesser extent, the prescribed role of the respondent and hence are not a function of the overall structural attributes of the organization of which the respondent was a member. In addition, whatever linkage might exist, between individual boundary-spanning behavior and individual perceptions of organicness, would have had little impact in this study, since the organicness score used was the mean of the scores obtained from all members of the individual's work unit while the boundary-spanning score was strictly a function of the individual's personal activity level. These variables were operationalized differently in order that, if a correlation were observed be-

Table 3

Factor Matrix for Perceived Environmental Uncertainty Measures			
Question	Factor Matrix		<i>h</i> ²
	1	2	
2	.75	−.05	.56
3	.71	−.03	.50
8	.47	.67	.67
9	.71	.24	.55
10	.69	−.51	.74
47	.37	.36	.22
48	.45	−.58	.54
Eigenvalues	2.6	1.2	

Table 4

Measures of Boundary-Spanning Activities

Excluding professional or occupational association, in an average week in the performance of your job—

1. How many hours do you attend formal meetings (committees, planning, etc.) (If not applicable, check here _____.)

a. Outside your work group but within the larger organization (includes regional offices)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 _____How many more?

b. Outside of your organization altogether (includes clients)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 _____How many more?

2. How many hours do you confer about your work in informal face-to-face conversation or telephone conversation with people of other organizations. This refers to informal and verbal contacts only. (If not applicable, check here _____.)

a. Outside your work group but within the larger organization (includes regional offices)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 _____How many more?

b. Outside of your organization altogether (includes clients)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 _____How many more?

3. How many times do you send or receive formal, written but work related communications in the form of reports or data from people in other organizations. Please include monthly or similarly scheduled reports in this response. (If not applicable, check here _____.)

a. Outside your work group but within the larger organization (includes regional offices)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 _____How many more?

b. Outside of your organization altogether (includes clients)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 _____How many more?

4. How many times do you send or receive informal, written communications in the form of reports, memos or the like from people in other organizations in regard to your work. (If not applicable, check here _____.)

a. Outside your work group but within the larger organization (includes regional offices)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 _____How many more?

b. Outside of your organization altogether (includes clients)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 _____How many more?

tween them, it would not be a consequence of our having measured the same variable two different ways.

Data Analysis

The individual subject was the unit of analysis. The 182 scores were organized in three contingency tables with approximately equal numbers of subjects in the marginals, as shown in Tables 6, 7, and 8. Since the data probably possessed only ordinal scale properties, they were first analyzed using the Goodman-Kruskal (1954) Gamma, γ , a nonparametric measure of association yielding values of $-1 \leq \gamma \leq +1$. This measure has an interpretation similar to that of a correlation coefficient, but is applicable to ordinal scale

Table 5

Factor Matrix for Boundary-Spanning Activity Items

Question	Factor Matrix			h ²
	1	2	3	
1.a.a.	.53	.23	.61	.7
1.a.b.	.55	.60	.36	.79
1.b.a.	.53	.25	-.55	.62
1.b.b.	.60	.56	-.39	.82
1.c.a.	.44	-.40	.12	.49
1.c.b.	.69	-.35	-.12	.61
1.d.a.	.68	-.47	-.06	.66
1.d.b.	.83	-.23	.09	.74
Eigenvalues	3.05	1.34	1.00	

Table 6

Relationship between Organicness and PEU*					
Organicness	Perceived Environmental Uncertainty (PEU)				Total
	34 and less	34–41	41–50	50–100	
Low	29	27	24	24	104
High	14	17	23	24	78
Total	43	44	47	48	182
$\gamma = .21 (p < .05)$					

• These data reflect approximately equal numbers of work groups in each category, but since there are unequal numbers of people in each work group, there are unequal *N*.

Table 7

Relationship between Organicness and Boundary-Spanning Activity*					
Organicness	Boundary-Spanning Activity				Total
	10 and less	10–28	28–62	62–404	
Low	39	29	21	15	104
High	5	15	25	33	78
Total	44	44	46	48	
$\gamma = .63 (p < .01)$					

• These data reflect approximately equal numbers of work groups in each category, but since there are unequal numbers of people in each work group, there are unequal *N*.

Table 8

Relationship between Boundary-Spanning Activity and PEU*					
Boundary-Spanning Activity	Perceived Environmental Uncertainty (PEU)				Total
	34 and less	34–41	41–50	50–100	
10 and less	19	9	12	4	44
10–28	11	11	10	12	44
28–62	6	12	11	17	46
62–404	7	12	14	15	48
Total	43	44	47	48	
$\gamma = .28 (p < .01)$					

• These data reflect approximately equal numbers of work groups in each category, but since there are unequal numbers of people in each work group, there are unequal *N*.

data. The data were further analyzed using a partial gamma, γ' , as in Davis (1967).

RESULTS

The three first-order γ values associated with the three pairings of variables (from Tables 6, 7, and 8) are shown in parentheses in Figure 1. Since all three values indicate positive associations between the respective variables (after adjusting for the reverse scoring on two of the vari-

ables), it is possible that the sign and magnitude of any one value could be simply the consequence of the sign and magnitude of the other two. In order to determine whether this were true, or if there really were underlying associations between the variables of each pairing, the second-order correlations are calculated and are shown within the square brackets of Figure 1.

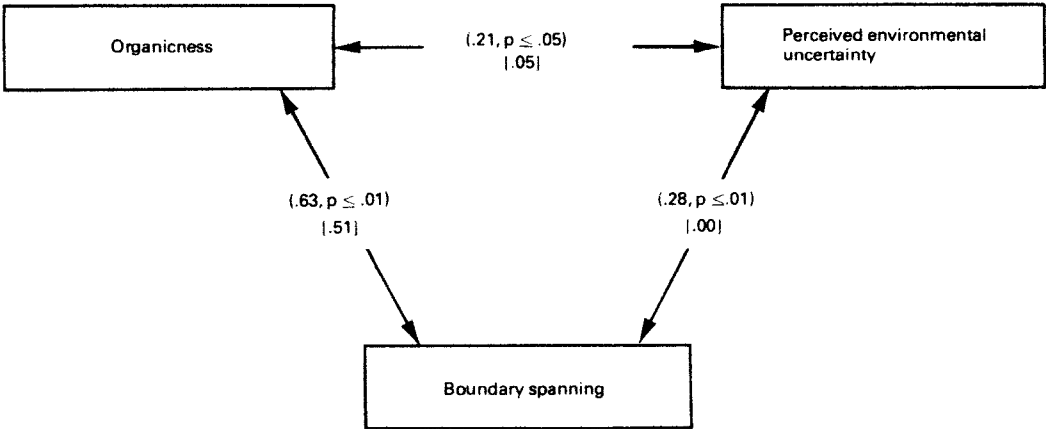


Figure 1. Relationships among boundary spanning, PEU and organicness.

Hypothesis 1 states that there will be a positive relationship between organicness and PEU. Table 5 and Figure 1 show that the observed relationship [$\gamma = .21$] is positive and marginally significant ($p < .05$), but does not exist [$\gamma' = .05$] after the variance due to boundary-spanning activity is removed. When only those individuals scoring in the upper quartile on the boundary-spanning measures are included, the PEU-organicness relationship is $\gamma = .36$ ($p < .01$), whereas when only those scoring in the lower three quartiles were included, γ is nonsignificant. These facts suggest that, at least at the boundary of the organization, the stronger association between organic structure and high PEU occurs primarily if not solely under the condition of high information processing activity.

Thus hypothesis 1 is supported subject to the strong qualification that the positive relationship appears to be influenced by boundary-spanning or information-processing activity that links the two variables of the hypothesis. This finding lends support to the notion that boundary spanning may have an impact on the relationship between PEU and structure and should be made a more explicit and studied component of contingency theory.

Hypothesis 2 states that there will be a positive relationship between organicness and boundary-spanning activity. Table 7 and Figure 1 show that hypothesis 2 is supported. The observed relationship ($\gamma = .63$) is positive and highly significant ($p < .01$) and exists ($\gamma' = .51$) even after the variance due to perceived environmental uncertainty is removed. Thus this hypothesis is supported without qualification.

Hypothesis 3 states that there will be a positive relationship between boundary-spanning activity and PEU. Table 8 and Figure 1 show that the observed relationship ($\gamma = .28$) is positive and highly significant ($p < .01$), but does not exist [$\gamma' = .00$] after the variance due to organicness is removed. Thus this hypothesis is supported subject to the strong

qualification that the positive relationship appears to be influenced by the organicness of the organization of which the individual is a member.

SUMMARY AND DISCUSSION

The results of this study suggest boundary-spanning activity might be usefully included as a component variable in an expanded form of contingency theory. In particular, the test of hypothesis 1 suggests that the frequency of boundary-spanning activity may mediate the association between structure and PEU.

This important finding may lead to the development of a richer theory of the relation between an organization and its environment. It introduces explicitly and empirically a variable that apparently has not been explicitly and empirically treated in the contingency theory literature; and, the variable introduced, boundary-spanning behavior, may allow for the integration of findings from the organizational behavior literature and the organization theory literature.

The finding that organicness and the frequency of boundary-spanning behavior are strongly associated is of considerable interest with respect to the second issue—the nature of the linkages among PEU, organicness, and frequency of boundary-spanning activity. Recall that this is not an association between specific role definitions—a measure of structure—and adherence to them—a measure of behavior, an association one would expect, in general, to be quite strong. Rather, this is the relation between an organizational attribute, organicness, and an individual behavior, communication frequency with external agents, a behavior that might be influenced by many factors, including PEU. The authors had reasoned that the association between these two variables was a consequence of PEU, that both would increase as PEU increased. The research results do not support this reasoning, as partialling out the effect of PEU reduced the association very little. This seems to be a provocative finding and suggests the need to propose and test alternative explanatory models.

The analysis performed here suggests that, even though PEU does not determine the association between organicness and boundary-spanning activity and even though the measures used to operationalize these variables do not necessarily imply an association, a strong relation does exist. Certainly more research on this matter is in order.

Pursuing further the issue of the nature and direction of the linkages among these three variables, the findings reported here suggest that boundary-spanning activity may intervene between or mediate the relationship between PEU and organization structure. Inasmuch as most studies in the literature relating to PEU utilize zero-order correlations, it has been logical for researchers to posit the PEU-influences structure (or outside in) reasoning that follows from an open systems theory of organizations, and which is consistent with current forms of contingency theory that deal with the organization-environment interface (see Kast and Rosenzweig, 1970; Utterback, 1971).

Findings from this study suggest an alternative to this position. The apparent relationship between PEU and organicness disappears when the frequency of boundary spanning is partialled out and the apparent relationship between PEU and boundary-spanning activity also disappears when organicness is partialled out. On the other hand, when PEU is partialled out of the relationship between structure and boundary-spanning activity, the relationship is not appreciably reduced. It appears that, on the basis of these results, structure influenced boundary-spanning activity to a greater degree than did PEU. In addition, it appeared that boundary spanning mediated the relationship between structure and PEU. This implies that inasmuch as structure is an administratively controlled variable, the amount of PEU admitted at the boundary of the organization is also administratively controllable. These results are complementary to previously found correlational relationships between structure and PEU only if an inside-out or structure-influences-PEU reasoning for the relationship is accepted.

While it is reasonable to assume the argument that structure influences boundary spanning which influences PEU, the data reported here might also support an argument that boundary spanning influences structure which influences PEU. If structure is a consequence of decisions based on information about the state of the environment which comes by way of boundary spanners, then a decision maker's PEU comes as a consequence of boundary spanning. As Leifer (1975) shows, boundary-spanning personnel also participate in making decisions under conditions of high PEU in which case boundary-spanning activity might influence structure, which then influences PEU. Further research might focus on the forces that influence the boundary spanner to perceive more or less environmental uncertainty.

In any case, the role of the boundary spanner as a mediator between environmental influences and organization structure is still supported and the importance to organization theory of the boundary-spanning function is underscored. It remains for additional research to tease out relations and conditions for those relations and to determine what effective boundary-spanning activity consists of. That is, is it effective for a boundary spanner in a mechanistic organization to report high amounts of uncertainty when it is clear the organization can not or need not deal with this uncertainty?

It should be reiterated that this study only pertained to the lowest levels of the fairly hierarchical and large institution studied, suggesting that the results reported here may be applicable only to the lowest organizational echelons. Certainly further research on the interrelationships among these variables at middle and upper levels of organizations is needed.

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